

1. Solve

$$x[n] = x[n-1] - \frac{1}{4}x[n-2]$$

with $x[-1] = 0$, $x[0] = 1$ using

(a) (25 points) z-transform techniques.

(b) (20 points) characteristic equations.

(a)

$$x[0] = x[-1] - \frac{1}{4}x[-2] \implies x[-2] = 4(0 - 1) = -4$$

$$X(z) = z^{-1}[X(z) + x[-1]z] - \frac{1}{4}z^{-2}[X(z) + x[-1]z + x[-2]z^2]$$

$$X(z) = z^{-1}X(z) - \frac{1}{4}z^{-2}X(z) + 1$$

$$X(z) = \frac{1}{1 - z^{-1} + \frac{1}{4}z^{-2}} = \frac{1}{(1 - \frac{1}{2}z^{-1})^2} = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2}$$

$$x[n] = \left(\frac{1}{2}\right)^n + n\left(\frac{1}{2}\right)^n = (n+1)\left(\frac{1}{2}\right)^n$$

(b)

$$r^0 = r^{-1} - \frac{1}{4}r^{-2}$$

$$1 - r^{-1} + \frac{1}{4}r^{-2} = 0$$

$$\left(1 - \frac{1}{2}r^{-1}\right)^2 = 0$$

$$r = \frac{1}{2} \quad \text{Multiplicity 2}$$

$$x[n] = a_1\left(\frac{1}{2}\right)^n + a_2n\left(\frac{1}{2}\right)^n$$

$$x[0] = a_1 = 1$$

$$x[-1] = 2a_1 - 2a_2 = 0 \implies a_2 = a_1 = 1$$

$$x[n] = \left(\frac{1}{2}\right)^n + n\left(\frac{1}{2}\right)^n = (n+1)\left(\frac{1}{2}\right)^n$$

2. (35 points) Find the general solution of

$$y^{(4)} + 8y'' + 16y = 0$$

$$r^4 + 8r^2 + 16 = 0$$

$$(r^2 + 4)^2 = 0$$

$$(r - 2j)^2(r + 2j)^2$$

$$r = \pm 2j \quad \text{Multiplicity } 2$$

$$y = Ae^{2jt} + Be^{-2jt} + Cte^{2jt} + Dte^{-2jt}$$

$$y = c_1\cos 2t + c_2\sin 2t + t(c_3\cos 2t + c_4\sin 2t)$$

3.

(20 points) Let $S_1 = \{r^n, nr^{n-1}, n(n-1)r^{n-2}, \dots, n(n-1)\dots(n-k+2)r^{n-k+1}\}$ and $S_2 = \{r^n, nr^n, n^2r^n, \dots, n^{k-1}r^n\}$ and

Prove or disprove: $\text{span } S_1 = \text{span } S_2$. (Note that the span of a set of sequences is the set of all sequences obtained as a linear combination of them.)

$$S_1 = \{r^n, nr^{n-1}, n(n-1)r^{n-2}, \dots, n(n-1)\dots(n-k+2)r^{n-k+1}\}$$

$$S_2 = \{r^n, nr^n, n^2r^n, \dots, n^{k-1}r^n\}$$

$$S_3 = \{r^n, nr^{n-1}, n^2r^{n-2}, \dots, n^{k-1}r^{n-k+1}\}$$

We will prove that $\text{Span } S_1 = \text{Span } S_2 = \text{Span } S_3$.

To see $\text{Span } S_1 = \text{Span } S_2$, observe that each term in

S_1 is a L.C.M. of the terms in S_2 :

$$\begin{aligned} \frac{1}{r^2} n(n-1)r^{n-2} &= n^2r^{n-2} - nr^{n-2} \\ &= \underbrace{1 \cdot n^2r^{n-2}}_{\sigma_1} + \underbrace{\left(-\frac{1}{r}\right)nr^{n-1}}_{\sigma_2} \end{aligned}$$

$$\begin{aligned} \frac{1}{r^3} n(n-1)(n-2)r^{n-3} &= n^3r^{n-3} - 3n^2r^{n-3} + 2nr^{n-3} \\ &= \underbrace{1 \cdot n^3r^{n-3}}_{\sigma_1} + \underbrace{\left(-\frac{3}{r}\right)n^2r^{n-2}}_{\sigma_2} + \underbrace{\left(\frac{2}{r^2}\right)nr^{n-1}}_{\sigma_3} \end{aligned}$$

etc. Hence they span the same set.

To see $\text{Span } S_2 = \text{Span } S_3$,

$$x \in \text{Span } S_2 \Leftrightarrow x = \alpha_1 r^n + \alpha_2 nr^n + \alpha_3 n^2 r^n + \dots + \alpha_k n^{k-1} r^n$$

$$\Leftrightarrow x = \alpha_1 r^n + \underbrace{\alpha_2 r}_{\beta_2} nr^{n-1} + \underbrace{\alpha_3 r^2}_{\beta_3} n^2 r^{n-2} + \dots + \underbrace{\alpha_k r^{k-1}}_{\beta_k} n^{k-1} r^{n-k+1}$$

$$\Leftrightarrow x = \alpha_1 r^n + \beta_2 nr^{n-1} + \beta_3 n^2 r^{n-2} + \dots + \beta_k n^{k-1} r^{n-k+1}$$

$$\Leftrightarrow x \in \text{Span } S_3$$